5. You are given a set $P = \{p_1, p_2, \ldots, p_n\}$ of n points in the two-dimensional plane. In addition, a set E of pairs of points is given. Two points p_i and p_j are said to be *adjacent* if $(p_i, p_j) \in E$. You are also given a positive integer k. The objective is to assign one of the k colors to every point such that no two adjacent points receive the same color.

   Design a recursive divide-and-conquer algorithm to determine whether such a coloring exists. Clearly describe the divide, conquer, and combine steps. Derive a recurrence for the worst-case running time and analyze it. (4 Mark)